MATH 110 — Week 10 — Solutions

Notes Chapter 18; Homework 10

HW10 P1.

Idea. Subtract earlier polynomial components; parity makes the odd mixed integrals zero.

1. Start with $u_1 = 1$, $u_2 = x$ and $u_3 = x^2 - \langle x^2, 1\rangle/\|1\|^2 = x^2 - 1/3$. They are orthogonal, and their squared norms are $2, 2/3, 8/45$.
2. Divide by those norms' positive square roots to obtain $e_1 = 1/\sqrt{2}$, $e_2 = \sqrt{3/2}\,x$, $e_3 = \sqrt{5/8}(3x^2 - 1)$. Proposition 87 gives an orthonormal basis with the original initial spans.

HW10 P2.

Idea. Orthogonality to a spanning list is a pair of homogeneous equations.

1. For (a, b, c, d) the equations are $a - b + c = 0$ and $2b + c = 0$. Thus $c = -2b$ and $a = 3b$, while b, d are free.
2. Therefore $W^\perp = \operatorname{span}((3, 1, -2, 0), (0, 0, 0, 1))$. The displayed vectors are independent by their second and fourth coordinates, so form a basis.

HW10 P3.

Idea. Use one integral equation and choose two independent solutions.

1. For $p = a + bx + cx^2$, $\langle p, x + 1\rangle = 2a + 2b/3 + 2c/3$. Its vanishing means $a = -(b + c)/3$.
2. Thus $p = b(x - 1/3) + c(x^2 - 1/3)$. These two polynomials are independent by their x and x^2 coefficients, so $x - 1/3, x^2 - 1/3$ is a basis of $W^\perp$.

HW10 P4.

Idea. Take one orthogonal complement of a sum, then take the complement again.

1. The sum identity gives $(U^\perp + W^\perp)^\perp = (U^\perp)^\perp \cap (W^\perp)^\perp = U \cap W$.
2. Take complements of both sides. Every subspace is finite-dimensional here, so the double-complement theorem gives $U^\perp + W^\perp = (U \cap W)^\perp$.

HW10 P5.

Idea. The orthogonal residual contributes the same squared error for every candidate in the subspace.

1. Decompose $v = u_0 + w$ with $u_0 \in U, w \in U^\perp$. For each j, $\langle v, e_j\rangle = \langle u_0, e_j\rangle$, so Proposition 86 gives $P_U v = u_0 = \sum_j \langle v, e_j\rangle e_j$.
2. For any $u \in U$, $v - u = (v - P_U v) + (P_U v - u)$ is an orthogonal sum. Hence $\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2$.
3. This proves the inequality, with equality exactly when $P_U v - u = 0$, that is $u = P_U v$.

HW10 P6 (optional).

Idea. A functional on $\mathcal{P}_1$ is determined by its values on $1, x$.

1. Write $p = a + bx$. Testing $f = 1$ gives $2 = \int_0^2 1\,dx = \langle 1, p\rangle = 2a$, so $a = 1$.
2. Testing $f = x$ gives $2 = \int_0^2 x\,dx = \langle x, p\rangle = 2b/3$, so $b = 3$. Thus $p = 1 + 3x$.
3. Linearity verifies the identity for every $f = c + dx$. Riesz uniqueness, or uniqueness of the two coefficient equations, proves uniqueness.

HW10 P7 (optional).

Idea. The norm bound forces the projector's range and kernel to be orthogonal, by testing one mixed scalar.

1. Orthogonal projection has the norm bound by Pythagoras, as proved in the projection-properties theorem.
2. Conversely assume the bound. Proposition 91 gives $V = \operatorname{range} P \oplus \ker P$. For $u \in \operatorname{range} P, w \in \ker P$, $P(u+tw) = u$ for every scalar t.
3. Put $c = \langle u, w\rangle$. If $w = 0$ then $c = 0$. Otherwise choose $t = -c/\|w\|^2$. Expanding gives $\|u+tw\|^2 = \|u\|^2 - |c|^2/\|w\|^2$.
4. The assumed norm bound requires this to be at least $\|u\|^2$, so $c = 0$. Hence $\ker P \subseteq (\operatorname{range} P)^\perp$. Their dimensions are both $\dim V - \dim \operatorname{range} P$, by rank-nullity and Proposition 89, so they are equal.
5. Therefore P projects onto its range along its orthogonal complement, which is precisely orthogonal projection.

LADR 6A Exercise 11.

Idea. Find the line component by its single inner product.

1. For $w = (1, 3)$ and $a = (1, 2)$, $\langle a, w\rangle = 7$ and $\|w\|^2 = 10$. Thus $u = 7w/10 = (7/10, 21/10)$.
2. Subtract to get $v = a - u = (3/10, -1/10)$, whose dot product with w is $3/10 - 3/10 = 0$. This gives all requested properties; uniqueness follows from the orthogonal direct sum.

LADR 6A Exercise 6.

Idea. A mixed inner product can be eliminated by a scalar choice, forcing a strict decrease unless it is zero.

1. If $\langle u, v\rangle = 0$, expanding gives $\|u + av\|^2 = \|u\|^2 + |a|^2\|v\|^2 \geq \|u\|^2$.
2. Conversely $v = 0$ makes the conclusion immediate. For $v \neq 0$, set $a = -\langle u, v\rangle/\|v\|^2$. Then $\|u + av\|^2 = \|u\|^2 - |\langle u, v\rangle|^2/\|v\|^2$.
3. The bound for this a forces the subtracted term to be zero, so $\langle u, v\rangle = 0$.